

AI-Powered Wildfire Early Detection and Alerting System

Using Multi-Source Remote Sensing & Spatio-Temporal Risk Prediction

PROACTIVE DISASTER MITIGATION



Case Study: Uttarakhand Forest Fire (2021)

Forest fires reported across multiple districts of Uttarakhand in April 2021.

Contributing Environmental Factors

- Significantly below-normal rainfall during winter season
- Low humidity levels across forest regions
- Dry vegetation conditions increasing fire susceptibility

Impact & Response

- Multiple active fire zones across Garhwal and Kumaon regions
- Deployment of Forest Department, Police, Fire Services, and NDRF teams
- No reported human casualties, but large forest areas affected
- Significant operational and economic impact due to fire spread

Detailed Report: [Ministry of Home Affairs Disaster Management Division](#)



The Problem We Cannot Afford to Ignore

The Threat

Wildfires devastate ecosystems, infrastructure, and communities — often under predictable conditions of low rainfall and humidity, yet we fail to act in time.

Current Limitations

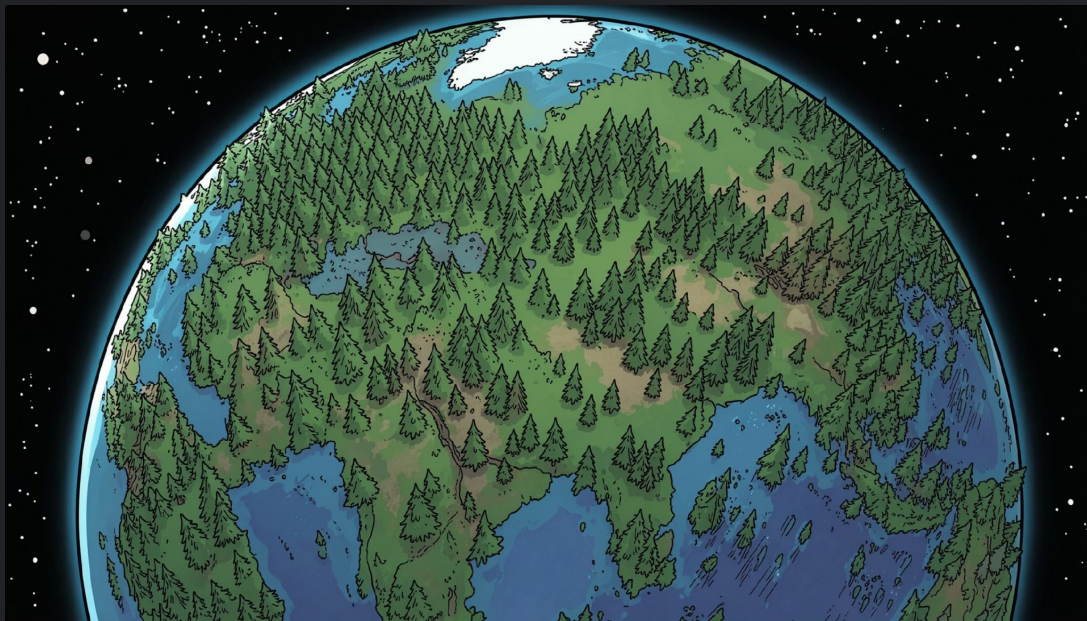
Management relies on **reactive measures**: post-ignition satellite hotspots, ground reporting, and emergency response after the damage begins.

The Critical Gap

There is a **critical lack** of intelligent systems capable of analyzing spatio-temporal data to predict wildfire risk *before* ignition occurs.

Our Plan: A Multimodal Approach

An AI-based spatio-temporal early warning system that fuses two complementary data streams to build a complete picture of evolving fire risk.



Satellite Imagery

Spatial-temporal data capturing vegetation health, land cover, and structural changes over time.

Environmental Data

Temporal atmospheric conditions — temperature, humidity, wind speed, and rainfall — that actively drive fire risk.

Core Objectives

1

Dataset Creation

Develop a multimodal dataset combining satellite imagery, environmental readings, and historical fire records.

2

Predictive Window

Forecast wildfire risk **24 to 48 hours** before a potential ignition event — enabling genuine early action.

3

Actionable Outputs

Generate geospatial wildfire risk heatmaps for vulnerable regions, accessible to decision-makers.

4

Automated Alerting

Implement an automated early warning alert system triggered for high-risk areas in real time.

5

Model Objectives

Primary Metrics: Accuracy & Recall ("Better Safe than Sorry") Secondary: Precision & F1-score



Who Does It Affect?



Authorities

Forest and wildlife protection departments, disaster management agencies (e.g., NDRF).



First Responders

Emergency response agencies and fire services requiring advance positioning of resources.



Vulnerable Populations

Communities living in or near forested, wildfire-prone regions who need timely evacuation warnings.



Strategic Planners

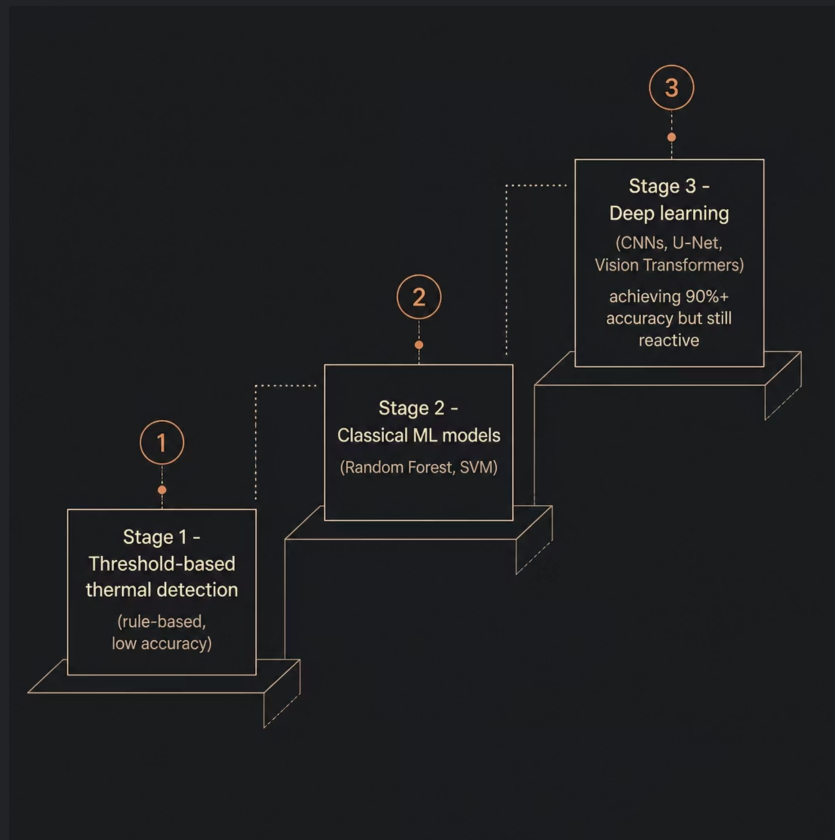
Environmental conservation organizations and climate policy makers shaping long-term land-use strategy.

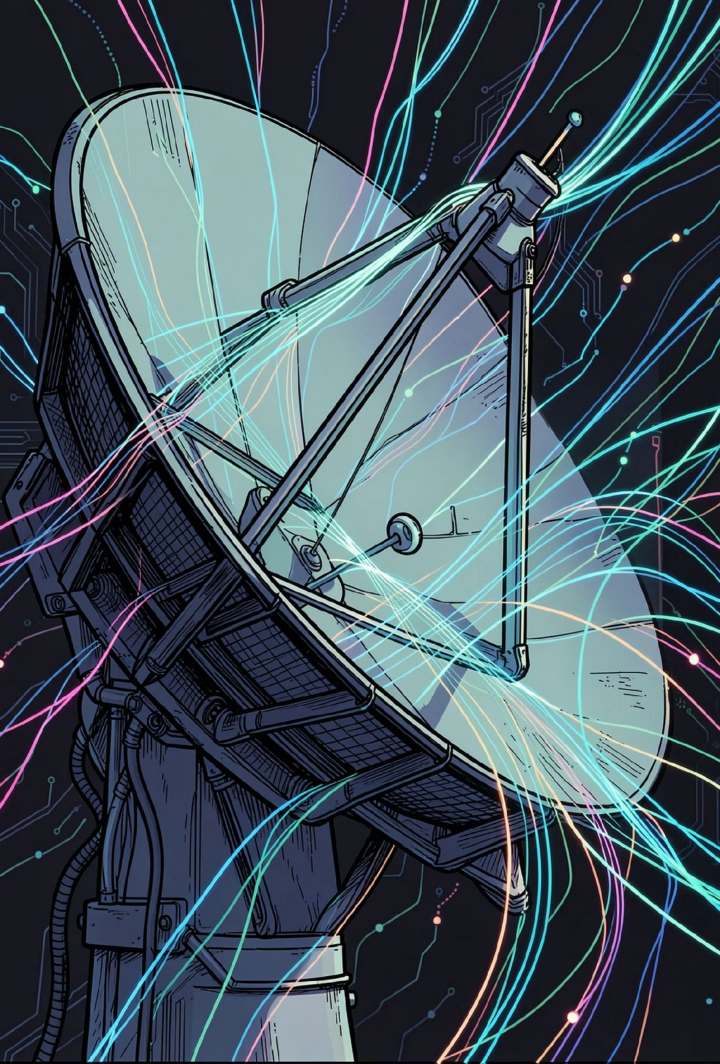
Literature Review & Current State

Where the Field Stands

Research has evolved from threshold-based thermal detection to deep learning architectures – CNNs, U-Net, and Vision Transformers – achieving **90%+ accuracy** in controlled settings. NASA FIRMS remains the operational backbone for real-world alerting via thermal hotspots.

- ▢ **The fundamental flaw:** Current systems are highly accurate but strictly reactive – they detect fires only *after* they start.





Identified Research Gaps

Single-Source Reliance

Most models use only optical or thermal imagery, ignoring weather, terrain, and atmospheric pollutants (aerosols).

Lack of Temporal Context

Models analyze single static images rather than evolving temporal trends over days.

No Risk Estimation

Systems detect active heat signatures rather than deriving pre-ignition risk scores.

Black-Box Models

Minimal use of Explainable AI (XAI) makes it hard for decision-makers to trust the alerts.

The Proposed AI Framework



Multi-Modal Fusion

Spatio-Temporal Learning

Attention Mechanisms

Explainable AI

This end-to-end pipeline unifies detection, risk scoring, and alert generation — replacing single-frame prediction with temporal reasoning and transparent, decision-ready outputs.

Methodology, Baselines & Evaluation

Baseline Models

Classical ML

Random Forest, SVM,
XGBoost

Deep Learning

U-Net, DeepLabV3+,
YOLOv8, Vision
Transformers

Evaluation Metrics

"Better Safe Than Sorry" — recall is prioritized to ensure no high-risk events are missed.

Industry Benchmarks

NASA FIRMS, Copernicus EMS, Google Earth Engine

Primary

Accuracy & Recall

Secondary

Precision, F1-score, IoU, ROC-AUC

Limitations & Conclusion

→ Probabilistic Nature

The model predicts risk levels, not the exact time of ignition.

→ Human Factors

Intentional arson or accidental human-induced fires cannot be directly observed from environmental data.

→ Data Boundaries

Geographic scope and performance are limited by satellite revisit times and image resolution.

✓ **Conclusion:** Despite limitations, this system represents a vital, proactive leap forward in global wildfire management – shifting the paradigm from reaction to prevention.



References

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2. [Deep Learning Approaches for Wildland Fires Using Satellite Remote Sensing Data \(2023\)](#)
3. [Deep Learning Approaches for Wildland Fires: Classification, Detection and Segmentation](#)
4. [Forest Fire Surveillance Systems: A Review of Deep Learning Methods \(2023\)](#)
5. [Uni-temporal Sentinel-2 Imagery for Wildfire Detection Using DL Semantic Segmentation \(2023\)](#)
6. [Comparative Analysis of CNN Architectures for Satellite-Based Forest Fire Detection \(2025\)](#)
7. [Copernicus Sentinel Documentation – Emergency Management Service](#)

References

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